COL780 Assignment 3: Image Classification and Attention Mechanisms

1. Results of Each Subtask (Cross-Entropy Baseline)

The initial phase of the assignment involved training four distinct architectures using standard Cross-Entropy loss. The models were evaluated on the test set for Accuracy, Macro F1-score, and Macro ROC-AUC.

Task	Architecture	Test Accuracy	Test Macro F1	Test Macro AUC
Task 1.1	Base ResNet-18	0.9844	0.9838	0.9995
Task 1.2	SE-ResNet-18	0.9811	0.9803	0.9995
Task 2.1	DeiT-3 Small	0.9826	0.9821	0.9993
Task 2.2	DyT-DeiT-3	0.9433	0.9400	0.9977

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Observation: The baseline ResNet-18 performed exceptionally well on this dataset. The Vision Transformer (DeiT-3) achieved highly competitive results, nearly matching the baseline ResNet-18. The DyT modification saw a noticeable drop in accuracy and F1 scores, though its AUC remained relatively high.

2. Ablation Study: Cross-Entropy vs. Focal Loss

To address class imbalances and hard-to-classify examples, an ablation study was conducted using Focal Loss.

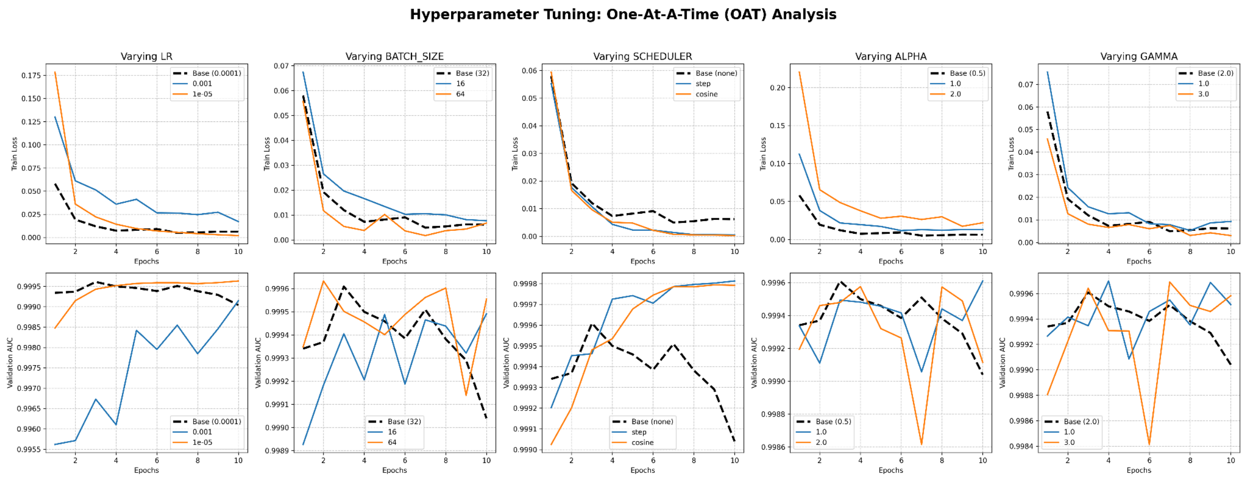
Hyperparameter Tuning: A grid search was performed to find the optimal Focal Loss parameters (alpha and gamma). The search revealed that the combination of alpha = 0.5 and gamma = 2.0 yielded the highest Validation AUC of 0.9998. These hyperparameters were then applied across all models.

Architecture	Loss Function	Test Accuracy	Test Macro F1	Test Macro AUC
ResNet-18	Cross-Entropy	0.9844	0.9838	0.9995
ResNet-18	Focal Loss	0.9833	0.9827	0.9996

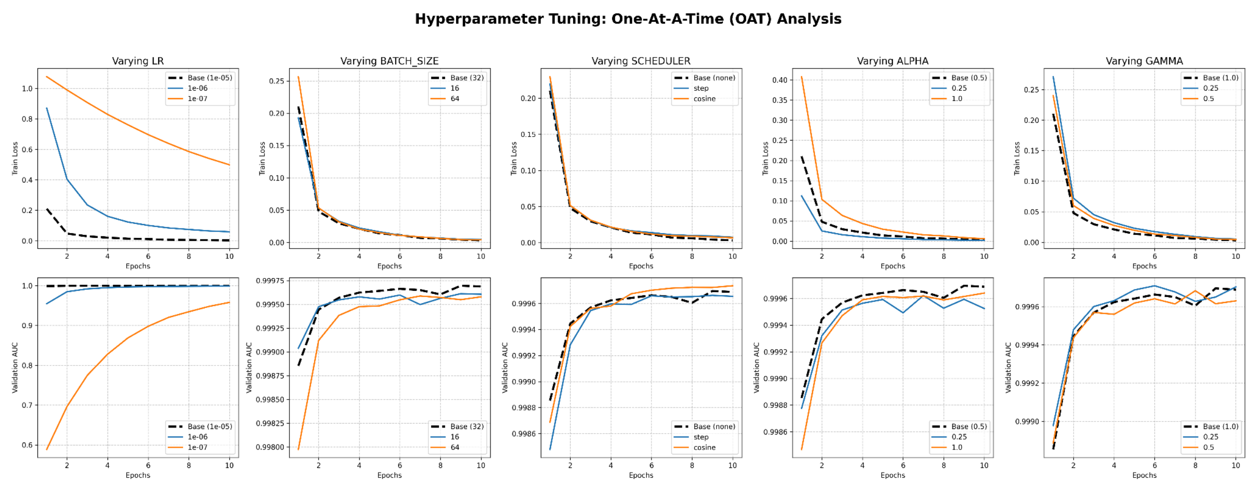
SE-ResNet-18	Cross-Entropy	0.9811	0.9803	0.9995
SE-ResNet-18	Focal Loss	0.9867	0.9859	0.9996
DeiT-3	Cross-Entropy	0.9826	0.9821	0.9993
DeiT-3	Focal Loss	0.9867	0.9859	0.9996
DyT-DeiT-3	Cross-Entropy	0.9433	0.9400	0.9977
DyT-DeiT-3	Focal Loss	0.9507	0.9484	0.9981

Impact Analysis: Focal loss provided a slight performance boost to the SE-ResNet-18, DeiT-3, and DyT-DeiT-3 architectures. However, the standard ResNet-18 saw a marginal decrease in accuracy and F1 score when switching from Cross-Entropy to Focal Loss. This suggests that for models already achieving near-perfect classification (like the baseline ResNet), dynamically down-weighting easy examples might slightly destabilize the learned boundaries.

3. Hyperparameter Tuning



Loss Curves and Validation AUC Curves - 1



Loss Curves and Validation AUC Curves - 2

Hyperparameter tuning was done on the following parameters :

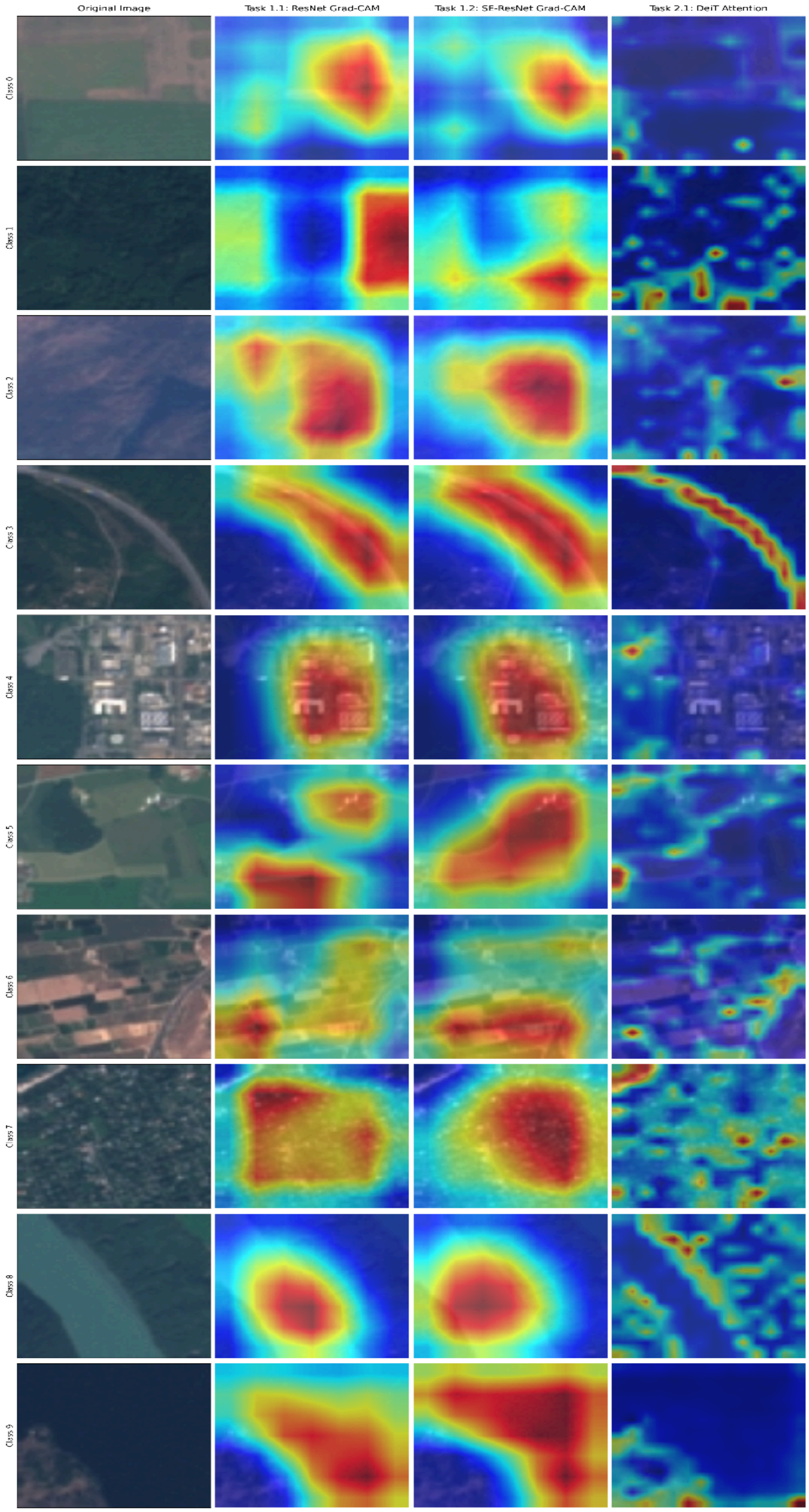
1. **Learning Rate**: *$1e^{-7}$, $1e^{-6}$, $1e^{-5}$, $1e^{-4}$, $1e^{-3}$*
2. **Batch Size:** *16, 32, 64*
3. **Scheduler:** *constant, step, cosine*
4. **Alpha:** *0.25, 0.5, 1.0, 2.0*
5. **Gamma:** *0.25, 0.5, 1.0, 2.0, 3.0*

After tuning, the best parameters thus obtained were :
Learning Rate = $1e^{-5}$
Batch size = *32*
Scheduler = *cosine*
Alpha = *0.5*
Gamma = *0.25*

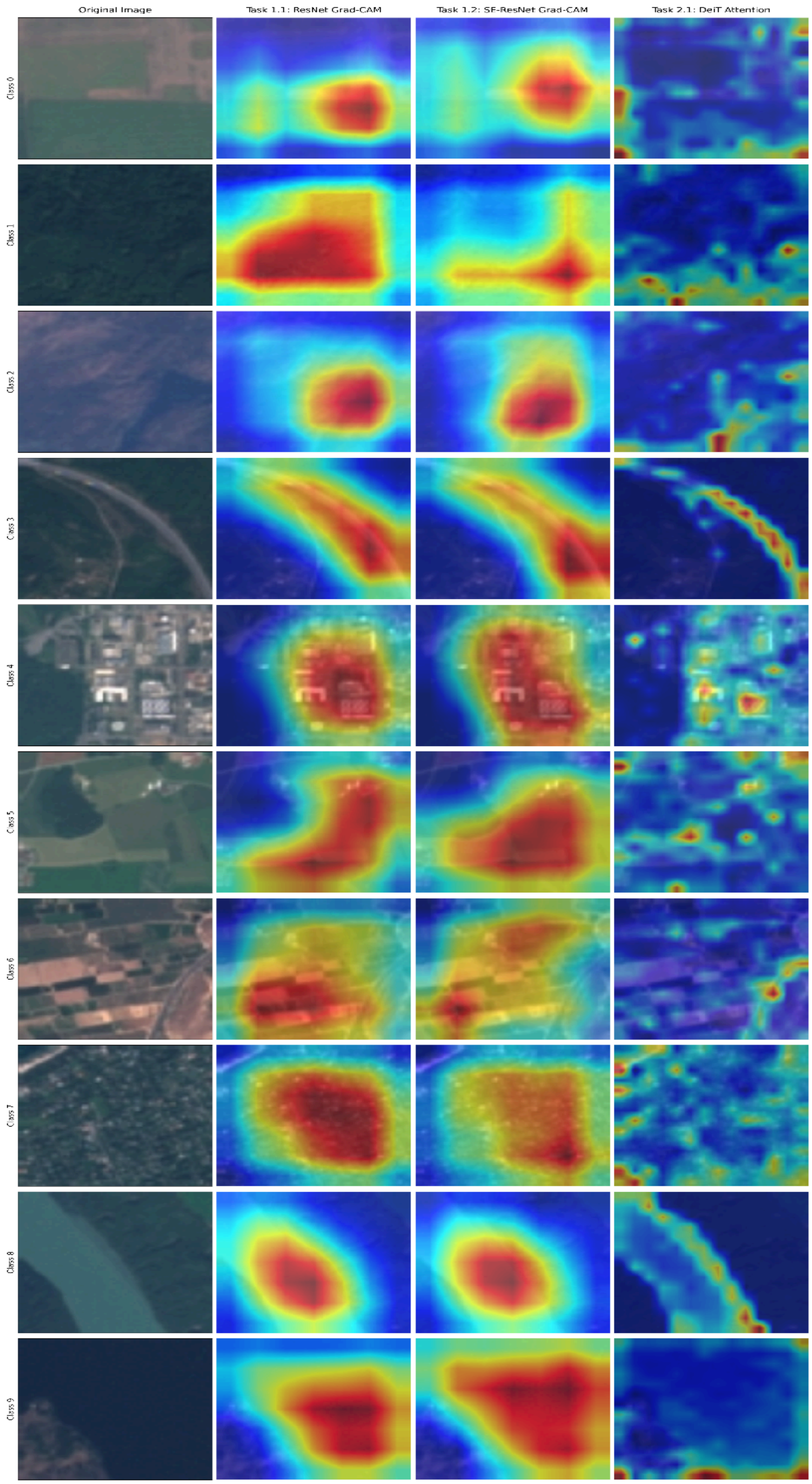
4. Analysis of Task 3 Images (Visualizations)

The visualizations successfully captured the fundamental differences in how Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) process spatial information.

- **CNNs (ResNet & SE-ResNet):** The Grad-CAM outputs produced smooth, continuous, blob-like heatmaps. This is due to the inherent inductive bias of convolutions, which look for local, continuous textures. For instance, in Class 8 (River), the CNNs highlighted the general mass of the water body. The SE-ResNet frequently demonstrated a slightly tighter or shifted focus compared to the Base ResNet, validating that the Squeeze-and-Excitation blocks successfully recalibrated channel importance to suppress noise.
- **Vision Transformers (DeiT-3):** The attention maps from the ViT were highly fragmented and patch-focused. Instead of broad textures, the DeiT prioritized high-contrast boundaries and structural edges. A prime example is Class 3 (Highway), where the ViT meticulously traced the sharp pixels of the road edges, whereas the CNN dropped a broad highlight over the general diagonal area. Similarly, for urban classes, the ViT attended to scattered individual structures rather than a continuous neighborhood block.



GRAD_CAM Visualisations for cross entropy loss



GRAD_CAM Visualisations for Focal Loss

5. Assumptions Made

- **Hyperparameter Transferability:** Due to significant computational constraints (GPU time limits on Kaggle), it was assumed that the optimal Focal Loss hyperparameters found via grid search on the baseline ResNet-18 would act as a sufficiently strong configuration for the remaining architectures.
- **Target Layer for SE-ResNet Grad-CAM:** It was assumed that the most informative layer for Grad-CAM extraction in the SE-ResNet was the final Squeeze-and-Excitation block (`se4`) prior to global average pooling, as this represents the feature maps immediately after channel recalibration.

6. Conclusions Drawn

1. **Architecture Efficacy:** Both standard CNNs and modern Vision Transformers are highly capable of tackling complex land-use classification tasks, both achieving over 98% accuracy.
2. **Mechanisms of Focus:** While CNNs and ViTs can arrive at the same correct prediction, their "reasoning" is fundamentally different. CNNs rely on regional textures, while ViTs are highly sensitive to structural edges and discrete patch relationships.
3. **Transformers without Normalization (DyT):** Replacing LayerNorm with Dynamic Tanh (DyT) in the DeiT architecture resulted in a noticeable performance degradation (dropping from ~98% to ~94% accuracy). While theoretically simpler, DyT may require longer training schedules, a different learning rate warmup, or specific architectural tweaking to match the stability of standard LayerNorm in this specific dataset.

7. Challenges Faced and Mitigations

- **CUDA Compatibility Issues:** Early on, training failed due to a hardware/software mismatch (Tesla P100 GPU architecture vs. newer PyTorch requirements). This was mitigated by dynamically altering the runtime environment to a T4 GPU, which fully supported the compiled PyTorch kernels.
- **Weight Loading Conflicts (.npz vs .pth):** A `RuntimeError` occurred when attempting to use standard `torch.load()` on numpy archive (`.npz`) files during the visualization task. This was solved by writing a custom helper function to read the `.npz` dictionaries and manually cast them back to PyTorch tensors before injecting them into the `state_dict`.
- **Transformer Attention Extraction:** Extracting the attention weights from the DeiT model initially failed with an `IndexError`. This occurred because modern PyTorch implementations utilize Scaled Dot Product Attention (SDPA), a C++ optimization that bypasses the standard `attn_drop` layer, causing the forward hooks to miss the data. The solution was to programmatically disable `fused_attn` on the final transformer block, forcing the model to route the matrices through the standard Python pathway for extraction.
- **Compute Limitations:** Running a comprehensive grid search for Focal Loss hyperparameters across all models would have exceeded available cloud compute limits. We tackled this by establishing a controlled experiment: tuning only on the baseline ResNet-18 and transferring the winning parameters to the other architectures.

8. References

1. **ResNet Architecture:** He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770-778.
2. **Squeeze-and-Excitation (SE) Blocks (Task 1.2):** Hu, J., Shen, L., & Sun, G. (2018). Squeeze-and-Excitation Networks. *Proceedings of the IEEE*

Conference on Computer Vision and Pattern Recognition (CVPR), 7132-7141.
3. **Data-efficient Image Transformers (DeiT-3) (Task 2.1):** Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A., & Jégou, H. (2021). Training data-efficient image transformers & distillation through attention. *International Conference on Machine Learning (ICML)*, 10347-10357.
4. **Transformers without Normalization / Dynamic Tanh (DyT) (Task 2.2):** Zhu, Y., et al. (2025). Transformers without Normalization. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
5. **Focal Loss (Ablation Study):** Lin, T. Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal Loss for Dense Object Detection. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2980-2988.
6. **Grad-CAM Visualizations (Task 3.1):** Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 618-626.
7. **PyTorch Image Models (`timm`) Library:** Wightman, R. (2019). PyTorch Image Models. *GitHub repository*. Retrieved from https://github.com/rwightman/pytorch-image-models

LINK TO THE WEIGHTS:

COL780_A3_weights